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Paper Id: 236682 Roll No.

B.TECH

(SEM VI) THEORY EXAMINATION 2022-23

ADVANCED MACHINE LEARNING

Time: 3 Hours *Total Marks: 100*

Note: Attempt all Sections. If require any missing data, then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) What is the role of Inductive Bias in ANN?

(b) How do you choose the number of hidden layers and nodes in a multi-layer perceptron?

(c) What is Bayesian learning?

(d) How can prior knowledge be incorporated into a Bayesian learning model?

(e) Explain the concept of pruning in decision tree models.

(f) Discuss overfitting and underfitting situation in decision tree learning.

(g) How Reinforcement Learning different from other types of machine learning?

(h) Explain the concept of a feedback network in reinforcement learning.

(i) What is a random forest?

(j) How does a random forest differ from a decision tree model?

SECTION B

2. Attempt any *three* of the following: 10x3=30

(a) Explain the concept of backpropagation and how it is used to train a neural network?

(b) Explain the concept of Markov Chain Monte Carlo (MCMC) methods in Bayesian learning, and how they are used for parameter estimation and model selection?

(c) How do decision trees compare to other machine learning algorithms, such as neural networks or support vector machines?

(d) Explain the various learning models for reinforcement learning.

(e) What is an ensemble in machine learning and how is it used to improve model performance?

SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) Explain the concept of gradient descent and how it is used in backpropagation?

(b) What are some real-world applications of multi-layer perceptron's and neural networks?

4. Attempt any *one* part of the following: 10x1=10

(a) How is Bayes theorem used in machine learning? How naive Bayes algorithm is different from Bayes theorem.

(b) What are some common challenges and limitations of Bayesian networks, and how can these be addressed?

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5. Attempt any *one* part of the following: 10x1=10

(a) How can decision trees be used for both classification and regression problems?

(b) List down the attribute selection measures used by the ID3 algorithm to construct a Decision Tree.

6. Attempt any *one* part of the following: 10x1=10

(a) Explain the role of Central Limit Theorem Approach for deriving Confidence Interval

(b) Describe Markov Decision Process in reinforcement learning.

7. Attempt any *one* part of the following: 10x1=10

(a) How do ensemble techniques compare to other machine learning algorithms, such as neural networks or support vector machines?

(b) What are some advantages and disadvantages of using ensemble techniques in machine learning?

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